

STEPH CURRY

2020-2021 NBA SEASON PERFORMANCE ANALYSIS

Data Analytics Capstone Project

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Dataset: stephcurry2020-2021.xlsx | 28 games, December 2020 – February 2021

30.1

Avg Points/Game
Season avg

5.0

Avg 3PM/Game
Three-pointers

49.2%

Avg FG%
Field goal %

15-13

Win-Loss Record
28 games played

1. About the Data

This analysis uses a game-by-game statistical log for Stephen Curry during a portion of the 2020–2021 NBA regular season. The dataset spans 28 games played between December 2020 and February 2021, covering 20 different opponents across home and away venues.

The 2020–2021 NBA season was played under a condensed schedule in the wake of the COVID-19 pandemic, resulting in a shortened season calendar and games played in limited-capacity or empty arenas for much of the year. As a result, some traditional home-court advantage effects may be muted compared to normal seasons.

Column Glossary

Readers unfamiliar with basketball statistics may find the following definitions useful:

Abbreviation	Full Name	Description
PTS	Points	Total points scored in the game
FGM/FGA	Field Goals Made/Attempted	All shot attempts excluding free throws
FG%	Field Goal %	FGM divided by FGA; measures overall shooting accuracy
3PM/3PA	Three-Pointers Made/Attempted	Shots made from beyond the arc (~23 ft)

Abbreviation	Full Name	Description
3P%	Three-Point %	3PM divided by 3PA; Curry's signature statistic
FTM/FTA	Free Throws Made/Attempted	Uncontested shots awarded after fouls
REB	Rebounds	Possession of the ball after a missed shot (OREB + DREB)
AST	Assists	Passes that directly lead to a made basket
STL	Steals	Taking the ball away from the opposing team
TO	Turnovers	Losing possession to the opposing team
P/M	Plus/Minus	Point differential for the team while Curry was on the court
COURT	Court Location	H = Home game, A = Away game
RESULT	Game Result	1 = Win, 0 = Loss (from the Warriors' perspective)

2. Problem Statement

The Golden State Warriors entered the 2020–2021 season without Klay Thompson, placing an extraordinary offensive burden on Stephen Curry. Understanding what conditions enable Curry to perform at his highest level — and how his individual performance translates into team wins — is not just an academic question. For coaching staff, it informs lineup decisions, shot selection strategy, and game planning. For analysts and front-office decision makers, it clarifies what aspects of Curry's game are most predictive of team success.

This analysis examines Curry's game-by-game performance across 28 games to answer the following: What factors drive Curry's scoring output, and how does his individual performance relate to the Warriors' win/loss outcomes? The goal is to identify measurable patterns that could inform strategic decisions about how the team is constructed around him and how his workload should be managed.

3. Business Questions

The following questions guide the analysis and are directly tied to the problem statement above:

- How does Curry's scoring and shooting efficiency vary by month throughout the season?
- Does playing at home vs. away affect his performance and the Warriors' win rate?
- Which shooting categories (FG%, 3P%, FT%) most strongly correlate with winning?
- How do his minutes played relate to points scored and overall team success?
- What does his plus/minus (P/M) reveal about his impact on game outcomes?
- How do his assists and turnovers relate to wins and losses?

4. Data Cleaning

The dataset arrived in a relatively clean state with no missing values across all 28 rows and 27 columns. However, several preprocessing steps were required before meaningful analysis could be performed.

Minutes Played (MIN)

The **MIN** column was stored as a timedelta (duration) object when read via pandas, reflecting the format in which Excel stored the values. To make this column usable for numeric analysis, it was converted to total minutes using:

```
df['MIN_num'] = df['MIN'].dt.seconds // 60
```

This produced an integer column representing total minutes on the court. After conversion, all values fell within a reasonable range for NBA playing time (approximately 25–44 minutes per game), confirming no data entry errors.

RESULT and COURT Columns

The **RESULT** column was already encoded as a binary integer (1 = Win, 0 = Loss), making it immediately usable for correlation analysis. The **COURT** column used 'H' and 'A' labels, retained as-is for groupby operations.

Remaining Quirks

One notable anomaly is the Toronto game in which Curry scored only 11 points on 12.5% shooting — yet the Warriors won. Conversely, his 62-point and 57-point performances both ended in either a win and a loss respectively. These outliers were retained because they reflect genuine game events and provide analytically meaningful variation. Removing them would distort the distribution and mask important nuance in the relationship between individual and team performance.

5. Analysis

Q1: How does scoring and efficiency vary by month?

Curry's performance showed a clear upward trajectory across the three months in the dataset. The chart below shows his per-game scoring for all 28 games, color-coded by win (blue) and loss (red), with a 5-game rolling average overlaid.

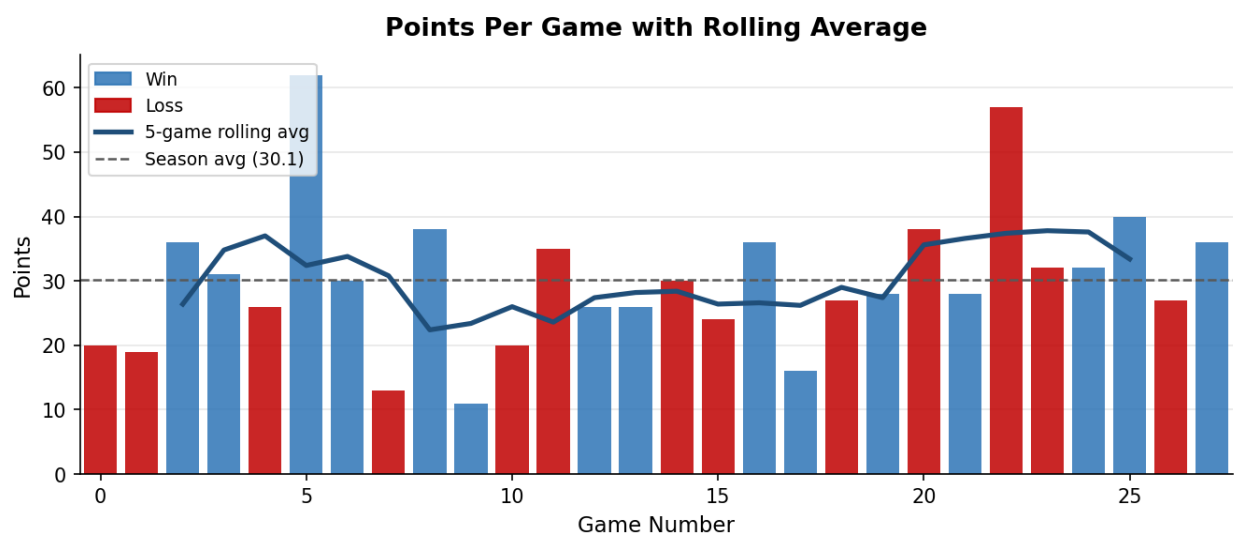


Figure 1: Points per game across all 28 games. Blue = win, red = loss. Rolling average shows the upward trend.

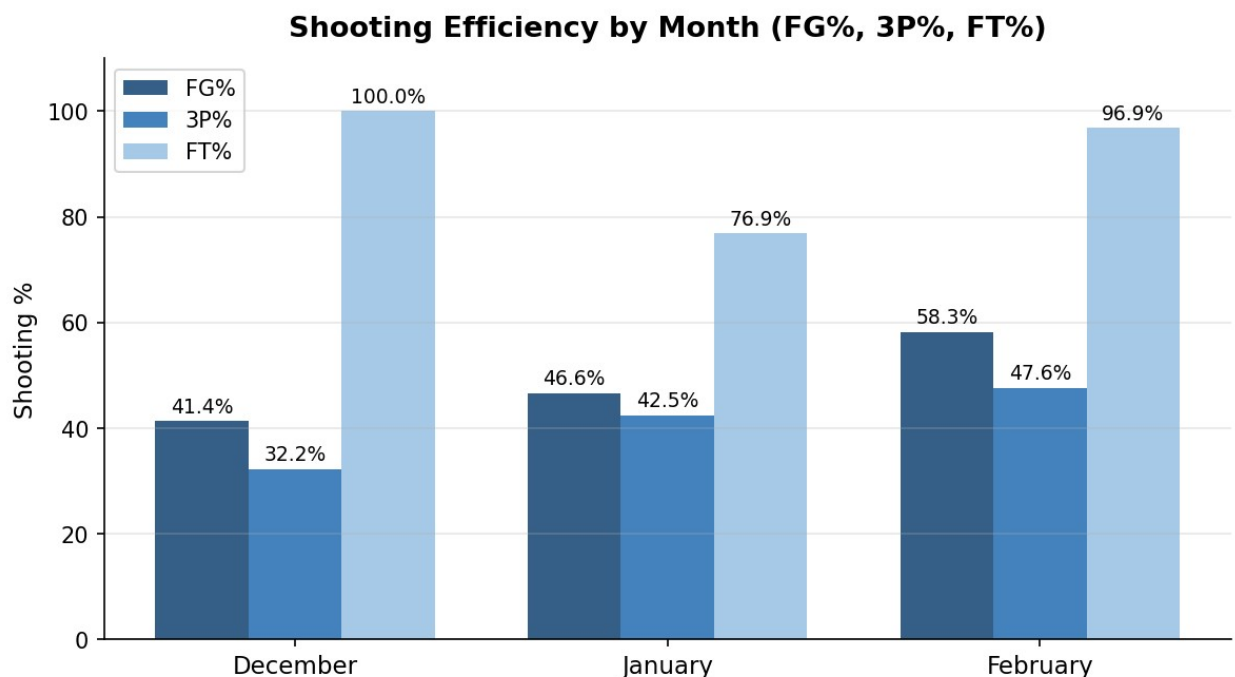


Figure 2: Average FG%, 3P%, and FT% by month. All three shooting categories improved from December to February.

Month	Avg PTS	Avg FG%	Avg 3PM	Avg 3P%	Win Rate
December	26.5	41.4%	3.5	32.2%	50%
January	28.0	46.6%	4.7	42.5%	56%
February	36.3	58.3%	6.4	47.6%	50%

February was Curry's most dominant month across virtually every shooting metric. His field goal percentage jumped to 58.3%, his three-point percentage climbed to 47.6%, and he averaged 36.3 points per game — nearly 10 more than in December. Notably, the win rate did not increase

proportionally with his scoring, indicating that individual hot streaks alone do not fully translate to team success.

Q2: Does home court affect performance and win rate?

The grouped bar chart below compares Curry’s key stats and the Warriors’ win rate in home versus away games.

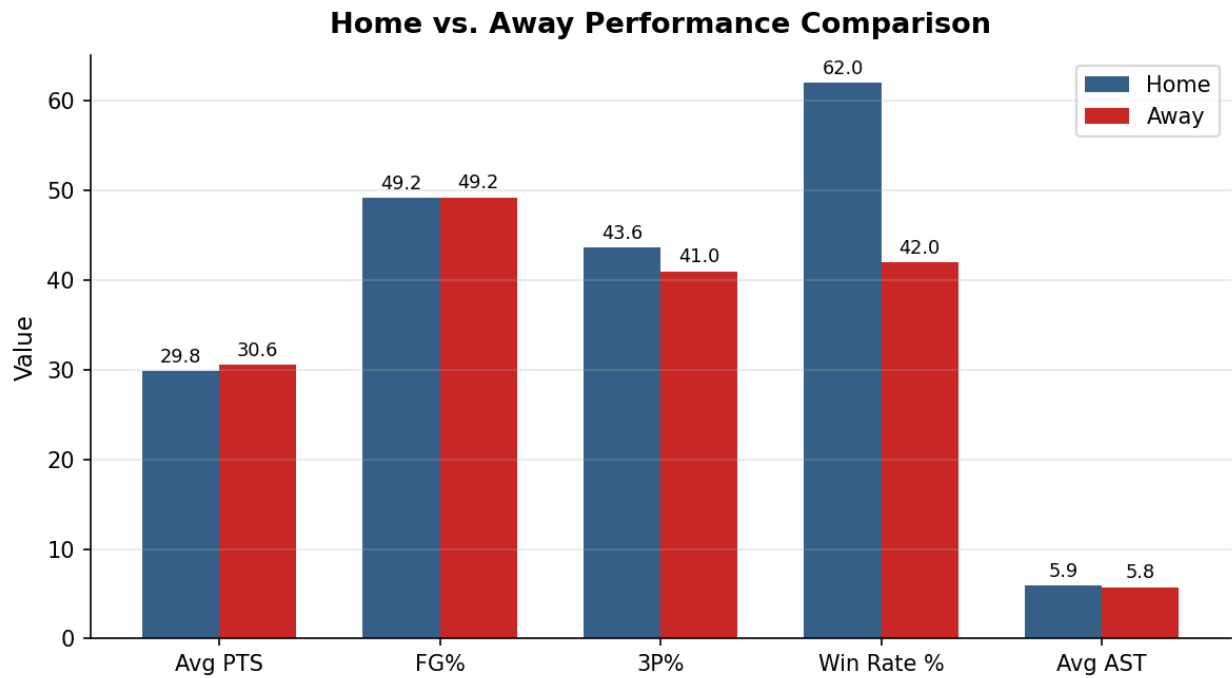


Figure 3: Home vs. away comparison across points, FG%, 3P%, win rate, and assists.

Location	Avg PTS	Avg FG%	Avg 3P%	Avg AST	Win Rate
Home (H)	29.8	49.2%	43.6%	5.9	62%
Away (A)	30.6	49.2%	41.0%	5.8	42%

Curry’s personal statistics were nearly identical at home and away. However, the Warriors’ win rate was markedly better at home (62%) than away (42%). This divergence suggests that while Curry performs consistently regardless of venue, the team around him performs significantly better at home. Home court advantage, even during a pandemic-shortened season with limited fans, still benefits the Warriors as a unit.

Q3: Which shooting categories correlate most with winning?

The horizontal bar chart below ranks each statistic by its correlation with a Warriors win.

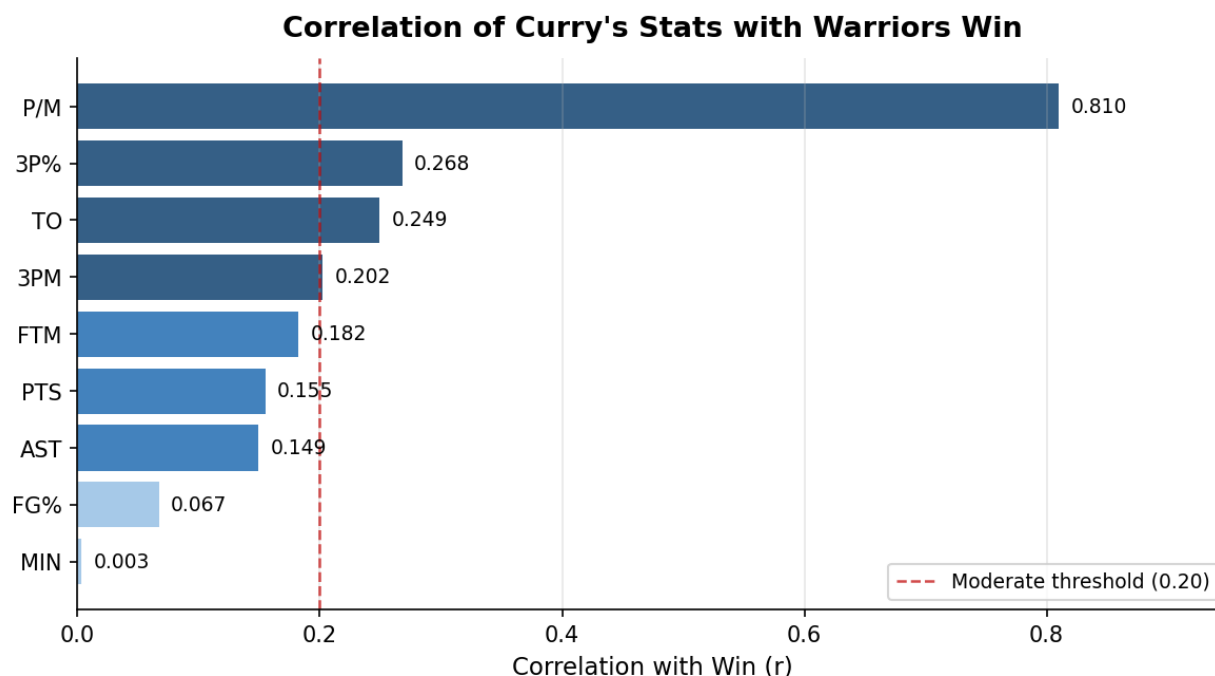


Figure 4: Pearson correlation of each statistic with the game result (1 = Win). Red dashed line marks the moderate threshold ($r = 0.20$).

Statistic	Wins Avg	Losses Avg	Correlation with Win
P/M (Plus/Minus)	+16.1	-14.0	0.81 (very strong)
3P%	46.4%	38.0%	0.27 (moderate)
TO (Turnovers)	3.67	2.69	0.25 (moderate)
3PM	5.5	4.5	0.20 (moderate)
FTM	5.5	4.2	0.18 (moderate)
AST (Assists)	6.1	5.5	0.15 (weak)
PTS	31.7	28.3	0.16 (weak)
FG%	50.0%	48.4%	0.07 (very weak)
Minutes	~9.7 hrs	~9.7 hrs	0.00 (none)

Plus/minus is by far the strongest predictor of a Warriors win ($r = 0.81$). Among individual shooting stats, **3-point percentage (0.27)** and **turnovers (0.25)** show the next strongest associations. Raw points scored and FG% show only weak correlations, suggesting that how efficiently Curry shoots the three — rather than overall shot volume — is more meaningful for team outcomes.

Q4: Do minutes played relate to scoring and team success?

Minutes played showed essentially zero correlation with the game result ($r = 0.00$). Curry averaged nearly identical minutes in wins and losses. The data does not support the idea that heavier playing time guarantees more wins — the quality of his shooting within those minutes matters far more than the quantity.

Q5: What does plus/minus reveal about Curry's impact?

The box plots below show the distribution of points, 3P%, and plus/minus across wins and losses, illustrating how each stat shifts between game outcomes.

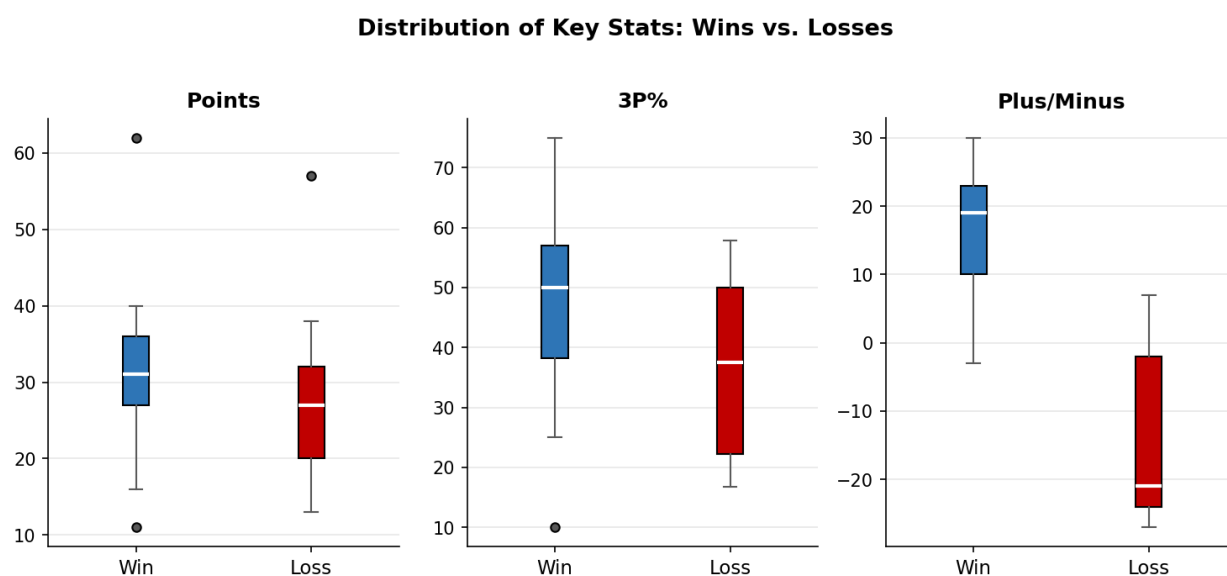


Figure 5: Distribution of Points, 3P%, and Plus/Minus in wins (blue) vs. losses (red). Plus/minus shows the clearest separation.

Plus/minus averaged **+16.1 in wins** and **-14.0 in losses** — a 30-point swing. The box plots confirm that plus/minus has the cleanest separation between outcomes, with almost no overlap between the win and loss distributions. Points and 3P% show moderate separation, but with considerable overlap, reflecting games where Curry scored well yet the team still lost.

Q6: How do assists and turnovers relate to outcomes?

Outcome	Avg Assists	Avg Turnovers	AST:TO Ratio
Win	6.1	3.67	1.66
Loss	5.5	2.69	2.04

A counterintuitive finding emerges: Curry actually averages more turnovers in wins (3.67) than in losses (2.69). One interpretation is that higher-turnover games reflect a more aggressive, attack-oriented style — Curry is making more high-risk, high-reward plays that, when they succeed, generate wins. Assist numbers also trend slightly higher in wins (6.1 vs. 5.5), consistent with a more

engaged, ball-moving performance. A passive, low-turnover Curry is not necessarily a winning formula for the Warriors.

6. Conclusion

This analysis examined 28 games of Stephen Curry’s 2020–2021 NBA season to understand what factors drive his individual performance and how that performance connects to Warriors wins and losses. Several clear patterns emerged.

Curry’s game improved substantially over the course of the season, with February representing a historic peak in scoring and efficiency. His personal numbers were consistent regardless of venue, yet the team performed significantly better at home — suggesting court location matters more for his supporting cast than for Curry himself.

Among all statistics examined, plus/minus proved to be the strongest predictor of team success ($r = 0.81$), followed by three-point percentage and turnover rate. Raw point totals and field goal percentage showed weaker connections to game outcomes, underscoring that efficiency from beyond the arc matters more than volume.

Three strategic takeaways emerge from this analysis: (1) maximizing Curry’s three-point opportunities and shooting efficiency is more valuable than ensuring he takes more shots overall; (2) the team around Curry needs to elevate its performance, particularly in away games, to convert his consistent individual output into wins; and (3) plus/minus serves as the most reliable in-game signal of whether a Curry performance is translating into team success.

Future analysis with a full-season dataset and opponent defensive ratings would allow for a more rigorous model and better generalize these findings beyond a 28-game window.

Appendix: Full Game Log

The table below shows the complete game-by-game statistical log used in this analysis.

Opp	Mo.	Court	PTS	FG M	FG A	FG%	3P M	3PA	3P%	AS T	TO	P/M	W/L
BRO	DEC	A	20	7	21	33.3	2	10	20.0	10	3	-23	L
MIL	DEC	A	19	6	17	35.3	2	10	20.0	6	2	-24	L
CHI	DEC	A	36	11	25	44.0	5	15	33.3	6	4	+3	W

Opp	Mo.	Court	PTS	FG M	FG A	FG%	3P M	3PA	3P%	AS T	TO	P/M	W/L
DET	DEC	A	31	9	17	52.9	5	9	55.6	6	8	+3	W
POR	JAN	H	26	9	20	45.0	4	12	33.3	5	1	-27	L
POR	JAN	H	62	18	31	58.1	8	16	50.0	4	5	+20	W
SAC	JAN	H	30	9	18	50.0	5	12	41.7	8	2	+26	W
LAC	JAN	H	13	5	17	29.4	1	6	16.7	5	0	-10	L
LAC	JAN	H	38	13	24	54.2	9	14	64.3	11	2	+13	W
TOR	JAN	H	11	2	16	12.5	1	10	10.0	6	5	+7	W
PAC	JAN	H	20	7	17	41.2	3	8	37.5	3	2	-2	L
DEN	JAN	A	35	14	23	60.9	5	11	45.5	4	7	-21	L
LAL	JAN	A	26	8	22	36.4	3	12	25.0	7	4	-3	W
SAS	JAN	H	26	10	17	58.8	4	8	50.0	7	4	+30	W
NYK	JAN	H	30	9	19	47.4	5	14	35.7	4	4	-16	L
UTH	JAN	A	24	9	18	50.0	5	10	50.0	7	1	-24	L
MIN	JAN	H	36	11	21	52.4	7	12	58.3	3	3	+19	W
MIN	JAN	H	16	6	17	35.3	4	11	36.4	8	2	+28	W
PHO	JAN	A	27	10	20	50.0	5	10	50.0	4	4	-23	L
DET	JAN	H	28	11	17	64.7	6	8	75.0	7	6	+26	W
BOS	FEB	H	38	12	21	57.1	7	14	50.0	8	3	+4	L
DAL	FEB	A	28	9	17	52.9	4	10	40.0	6	2	+19	W
DAL	FEB	A	57	19	31	61.3	11	19	57.9	5	3	+7	L
SAS	FEB	A	32	10	17	58.8	6	11	54.5	6	1	+3	L

Opp	Mo.	Court	PTS	FG M	FG A	FG%	3P M	3PA	3P%	AS T	TO	P/M	W/L
	B												
SAS	FEB	A	32	11	20	55.0	4	10	40.0	2	1	+15	W
ORL	FEB	H	40	14	26	53.8	10	19	52.6	5	6	+16	W
BRO	FEB	H	27	10	17	58.8	2	9	22.2	5	4	-26	L
CLE	FEB	H	36	13	19	68.4	7	11	63.6	6	1	+20	W